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The good, the bad, and the algorithm: The impact of generative AI on cybersecurity

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ABSTRACT

Generative Adversarial Networks (GANs) are emerging as a transformative technology in cybersecurity, presenting both opportunities and challenges in enhancing defensive and offensive strategies. This paper explores the impact that Generative Artificial Intelligence (AI) has on cybersecurity, focusing on its application in the field of network and web security. Current research reveals robust defensive approaches; however, there remains a significant gap in the application of Generative AI to develop advanced attack scenarios capable of bypassing existing defense mechanisms. Our work attempts to fill this gap and spreads awareness regarding a potential exposure of Neural Network (NN)-based Intrusion Detection Systems (IDSs) against AI-enhanced attacks. Unlike conventional approaches that focus on Input Perturbation, Data Poisoning, or Spoofing, we propose a novel offensive strategy called Attack Obfuscation. This strategy leverages Conditional GANs (CGANs) to conceal genuine attacks by injecting synthetic traffic designed to deceive NN-based IDS. The experimental investigation validates the proposed approach against three distinct datasets and different typologies of attacks, managing to successfully deceive the IDS.

1. Introduction

Artificial Intelligence (AI) technologies, specifically Generative Adversarial Networks (GANs), are reshaping cybersecurity through their applications in both defensive and offensive strategies. The research community explored multiple defensive applications of Generative AI in security systems [1–4]. Within Intrusion Detection Systems (IDS), GANs facilitate the generation of representative traffic patterns that enhance training efficacy and anomaly detection capabilities. Through adversarial example generation, Generative AI enables the simulation of sophisticated attack vectors, thereby strengthening system resilience. In malware analysis applications, the generation of synthetic samples provides deeper insights into malicious behavior patterns, contributing to the development of robust countermeasures. The technology demonstrates particular utility in phishing detection through the generation of training datasets comprising suspicious communications and URLs, thus improving classification accuracy. Moreover, it enables the automation of threat response mechanisms and security policy optimization through systematic network traffic analysis.

However, substantial challenges persist, particularly regarding the exploitation potential of Generative AI. A critical consideration is the dual-nature applicability of these capabilities. While security researchers employ Generative AI for defensive purposes, malicious actors may leverage similar techniques to develop sophisticated evasion methodologies that circumvent current protection mechanisms [5,6].

In this work, we provide the following contributions:

- Systematic Literature Review (SLR): We comprehensively examine Generative AI's role in cybersecurity, focusing on network and web security applications. Our analysis categorizes defensive strategies into *Adversarial Training*, *Dataset Balancing*, *Data Augmentation*, and *Data Obfuscation*. For offensive AI applications, we classified existing approaches into: *Input Perturbation*, *Data Poisoning*, and *Spoofing*.
- Novel Attack Obfuscation Strategy: We address a critical gap in attack concealment methodologies by developing and evaluating a Conditional GAN (CGAN)-based approach. Our method subverts a Neural Network (NN)-based IDS, enabling the success of a range of web and network attacks without altering their actual signatures. Specifically, the successfully concealed attacks include Man-In-The-Middle (MITM), three Mirai variants, and SQL In-

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jection. We further validate this approach using Zero-day attack datasets to assess its effectiveness against novel attack patterns.
- Comprehensive Model Stability and IDS Validation: We establish our method's reliability through statistical validation using one-way ANalysis Of VAriance (ANOVA), confirming NN-IDS model stability across multiple trials. We extend our evaluation to Random Forest (RF)-based IDS, demonstrating consistent results across different architectures.

The remainder of the paper is organized as follows: Section 2 presents the SLR methodology. Section 3 and Section 4 examine defensive and offensive AI applications respectively. Section 5 discusses identified research gaps. Section 6 details our novel attack obfuscation method. Section 7 analyzes experimental results, and Section 8 ends the paper with closing remarks.

2. Research method

This Systematic Literature Review (SLR) was conducted to identify, select, and analyze studies to answer the defined research questions (RQs), following the guidelines suggested in [7]. The methodology adopted for the SLR is detailed in the following subsections. Initially, the RQs were defined, along with relevant keywords for the search process. After defining inclusion and exclusion criteria and selecting relevant papers, the extracted data is evaluated and discussed in Sections 3 and 4.

2.1. Research questions and motivation

To provide a comprehensive overview of recent literature on the use of Generative AI in cybersecurity, the following research questions (RQs) were defined:

- RQ1: How does Generative AI contribute to enhancing cybersecurity strategies, particularly in the areas of intrusion detection and network protection?
- RQ2: What are the primary methods and techniques employed by attackers leveraging Generative AI to develop sophisticated cyber threats and evade traditional defense mechanisms?
- RQ3: Which possible unexplored usage of Generative AI might represent a concern when developing intrusion detection strategies?

The research questions (RQs) for this study are motivated by several key considerations. First, there is a compelling need to examine the existing literature to identify state-of-the-art approaches that leverage GANs for defensive purposes, such as intrusion and anomaly detection. Understanding both the strengths and limitations of these approaches is crucial to developing more robust and effective defense mechanisms (RQ1).

Second, as adversaries continuously adapt their tactics, it is essential to anticipate and address potential vulnerabilities that may arise from the integration of Generative AI in cybersecurity. This requires a proactive exploration of offensive strategies that adversaries might use, such as generating adversarial samples designed to bypass AI-based intrusion detection systems (RQ2).

Finally, this study aims to identify gaps in the current research, laying the foundation for the proposed experimental approach (RQ3).

2.2. Search process, study selection and quality assessment

To identify papers relevant to the formulated RQs, keywords were defined and combined with search operators to create the following query: "generative adversarial network" OR "conditional generative adversarial network" AND ("cybersecurity" OR "network security" OR "intrusion detection" OR "attack evasion" OR "obfuscation"). The research has been performed among the title, keywords, and abstract of the studies, by exploring the following databases: Web of Science,[2] Scopus,[3] and IEEE Xplore.[4] Inclusion and exclusion criteria were defined by (a) excluding any work published before 2020 for an up-to-date analysis, (b) excluding any duplicate studies and preliminary versions of the same work, (c) excluding works not written in English, (d) excluding reviews, surveys, and works that were not research papers, (e) excluding paper that were not accessible, and (f) excluding papers that were out of scope. The inclusion criteria were the opposite of the exclusion ones. The process led to the selection of the final set of 67 articles, as shown in Tables 1 and 2.

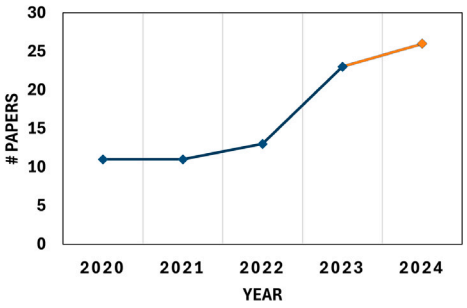


Fig. 1. Number of works per year.

Once identified, the papers were classified into two categories, to discriminate between defensive and offensive usage of Generative AI, answering RQ1 and RQ2. Then, each category was split in additional ones, according to the major approaches identified and as detailed in Sections 3 and 4, respectively. This classification was of support in answering RQ3, as shown in Section 5. Notably, some studies address both aspects by performing attacks and subsequently testing defense mechanisms against them. In such cases, the works have been classified in accordance with the main scope of the proposed research. Fig. 1 illustrates the number of papers published each year, demonstrating a growing interest in the topic over time. Fig. 2 depicts the trend of Defensive and Offensive approaches throughout the years, showing that their proportions have remained relatively stable. Both graphs include projections for 2024, indicated by a different color. These projections are based on a comparison between the first trimester of 2023 and the first trimester of 2024, assuming a consistent trend. It is important to note that this estimate is based on a small sample size, and the actual values might differ, as they also depend on various factors such as technology trends and other external influences. Figs. 3 and 4 show the percentage of Defensive and Offensive approaches according to the classification. Fig. 5 shows the number of authors per paper, where it can be noted that the majority of the works were written by groups ranging from 3 to 6 authors. Lastly, Fig. 6 is a graphical representation of the most recurring keywords extracted from the selected papers. The font size describes how much the keyword is indexed: as expected, the most indexed keywords are "network", "detection", and "attack". Tables 1 and 2 contain a classification of the research works as previously explained.

3. Defensive strategies using Generative AI

As shown in Table 1, most findings in the literature focus on the defensive applications of Generative AI. This section explores the key strategies identified in the literature.

[2] https://www.webofscience.com/wos/woscc/basic-search
[3] https://www.scopus.com/search/form.uri?display=advanced
[4] https://ieeexplore.ieee.org/search/advanced

Table 1
Summary of defensive strategies using generative AI.

Defensive strategy	Description	Key application domains	Literature findings
Adversarial Training	Techniques to enhance model robustness against adversarial attacks by training with adversarial samples generated via GANs.	Network IDS, Malware detection, DDoS detection, Ransomware detection	[8], [9],[10], [11], [12] [13], [14], [15], [16], [17], [18], [19], [20], [21]
Dataset Balancing	Techniques like oversampling, undersampling, and GAN-based methods to address class imbalance in datasets.	Network intrusion detection, ICS environments	[22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47], [48], [49], [50]
Data Augmentation	Generating synthetic data to expand training datasets and improve model generalization and robustness.	IoT security, Malware detection, Cyber threat analysis	[51], [52], [53], [54], [55], [56], [57], [58], [59]
Data Obfuscation	Obscuring sensitive data using GANs to prevent malicious actors from extracting private information.	Network traffic, Location data, Smart grid systems	[60], [61], [62]

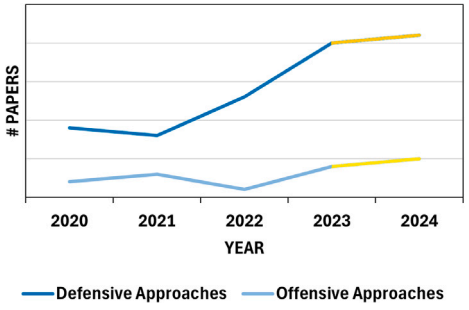


Fig. 2. Trend of Defensive vs Offensive approaches throughout the years.

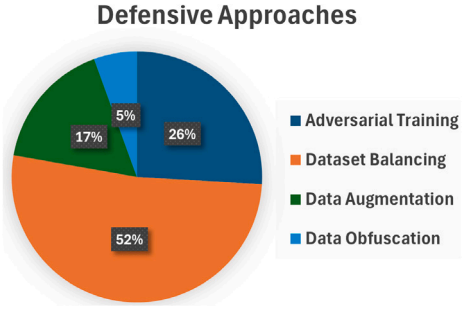


Fig. 3. Percentage of the different Defensive Approaches identified.

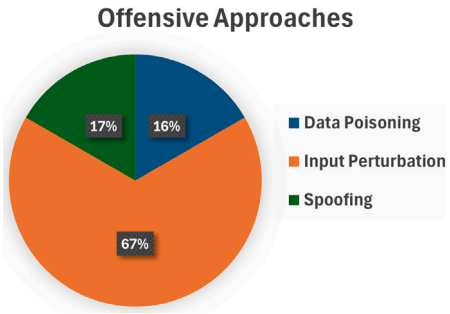


Fig. 4. Percentage of the different Offensive Approaches identified.

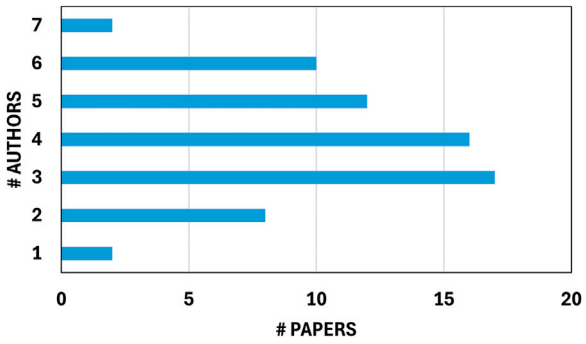


Fig. 5. Collaborative groups of authors.

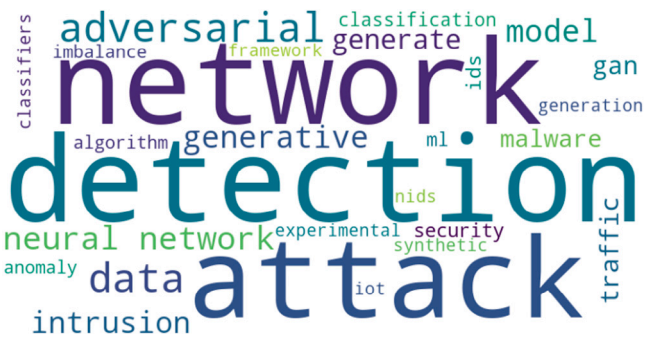


Fig. 6. The extracted Top Keywords cloud.

3.1. Adversarial Training

In Machine Learning (ML), Adversarial Training involves training a model on adversarial samples specifically crafted to deceive it. This exposure improves the model's ability to generalize and enhances its resilience to similar attacks.

Numerous studies propose approaches to achieve this across diverse scenarios. For example, [13] introduces a novel defensive mechanism based on a GAN framework to enhance the robustness of end-to-end communication systems against adversarial attacks. Zhou et al. [17] address limitations in Intrusion Detection Systems (IDS) that struggle with detecting unknown threats. They propose a detection algorithm leveraging GANs and evolutionary computation to derive unknown threat characteristics from known threats.

Similarly, [18] explores intrusion detection and prevention, specifically targeting evasion techniques employed by attackers. By generating adversarial instances of network traffic using GANs, this study demonstrates how ML-based IDS can be trained and improved to counter such evasion attempts. Zhang et al. [8] further extend this concept by employing Adversarial Training to optimize cross-site scripting (XSS) detection. Their approach uses GANs alongside a Monte Carlo tree search (MCTS) algorithm to enhance detection rates in the presence of XSS samples.

Malware detection is another critical application area. For instance, [9] proposes a GAN-based method to classify unobfuscated and obfuscated malware. The GAN visualizes local features and generates local images of obfuscated malware by learning from global and local features of unobfuscated malware. A similar GAN-enhanced methodology is explored in [20], which integrates both signature-based and AI-based detection techniques within the Anti-Malware System (AMS) architecture to detect unknown malware.

In the domain of Smart Grids, Siniosoglou et al. [10] develop an IDS utilizing an Autoencoder GAN architecture to improve the classification of operational anomalies and cyberattacks. Similarly, Adversarial Training shows promise in addressing Denial of Service (DoS) and Distributed DoS (DDoS) attacks. Novaes et al. [11] present a GAN-based defense mechanism for Software-Defined Networking (SDN) systems, enhancing robustness against adversarial attacks. Comparable approaches, such as those in [12,16], focus on training Network IDS (NIDS) using adversarial DDoS samples generated by GANs. While these studies share a common objective of improving NIDS performance, they differ in methodology, datasets, evaluation metrics, and performance outcomes, offering diverse insights into the applicability of Adversarial Training in real-world scenarios.

However, Adversarial Training is not without drawbacks. Choi et al. [14] highlight the risk of degraded accuracy on legitimate samples. To mitigate this issue, they propose a novel GAN-based defense mechanism where the generator learns to mimic the target model's vulnerabilities and adjusts its parameters using a combined loss function.

Other innovative approaches include a Transferred GAN (TGAN)-based IDS for detecting unknown or variant ransomware attacks, as proposed in [15]. This IDS employs dual GANs—one for generating adversarial samples and the other for detecting attacks—while maintaining the ability to classify normal samples through a construction loss function. A similar system is introduced in [19], which combines advanced techniques such as 3D cube-based secure authentication, modified honeycomb network partitioning, reinforcement learning-based routing, and a bidirectional GAN (Bi-GAN) for intrusion detection in software-defined wireless sensor networks (SDWSNs).

Lastly, Li et al. [21] present a unique GAN-based Adversarial Training framework for NIDS. By emphasizing the interaction between the Generator and Discriminator, their approach generates robust and diverse adversarial samples, further strengthening the NIDS against evolving threats.

3.2. Dataset Balancing

Class imbalance is a common issue in machine learning datasets, where certain classes are underrepresented compared to others, leading to biased and inaccurate classification results [63].

Dataset Balancing addresses this by oversampling the minority class, undersampling the majority class, or using advanced techniques like the Synthetic Minority Over-sampling Technique (SMOTE) to generate synthetic samples. These methods help improve model performance by reducing bias toward the majority class. In the literature, this issue is extensively tackled using GANs [23,28,30,33,35,37,38,41,47,49], and CGANs [24–27,31,32,39,40,43,46,48]. These approaches aim to generate balanced datasets for anomaly and intrusion detection tasks. Differences among these methods primarily lie in architectural variations, such as the number of generators used.

Some studies focus on specific applications of CGANs. For instance, [22] uses CGANs to generate synthetic samples that mimic the distribution of real XSS attack scenarios. The augmented data is then used to train a new model to demonstrate the validity and reliability of the fake samples. Xie et al. [29] propose a multi-label detection model based on deep learning. The approach utilizes Wasserstein Generative Adversarial Networks with Gradient Penalty (WGAN-GP) for data enhancement and to address the issues related to class imbalance. Similarly, in [34], a novel multi-module integrated intrusion detection system, GMM-WGAN-IDS, is proposed. This system combines stacked autoencoder-based feature extraction, clustering with Gaussian Mixture Model (GMM), and WGAN-GP for imbalance processing, followed by classification using a Convolutional NN (CNN) and long short-term memory (LSTM) model. In [36], a Data Augmentation technique based on WGAN-GP is presented to enhance the recognition accuracy of sparse attacks in network intrusion detection. The study evaluates various sparse data generation methods and implements a two-stage fine-tuning algorithm based on WGAN-GP to optimize classification accuracy. Additionally, in [42], a novel data augmenting approach called GraphCWGAN-GP is introduced. This approach converts traffic data into grayscale images and utilizes conditional constraints and a new distance metric in GAN to augment minority class data. A different approach, called Conditional Wasserstein Variational Autoencoders with GAN (CWVAEGAN), is proposed by He et al. [44] to generate minority class samples in unbalanced datasets. CWVAEGAN addresses the class-imbalance problem in network traffic datasets by transforming the data and utilizing an improved VAEGAN to generate minority class samples. An intrusion detection system, CWVAEGAN-1DCNN, is established, integrating CWVAEGAN with One-dimensional convolutional NN (1DCNN). Comparably, Cai et al. [45], propose a 1D CWGAN to balance normal and attack samples in industrial control systems (ICSs) network environments, significantly improving classification performance. An IDS combining deep networks and hybrid sampling for Dataset Balancing is presented in [50], featuring Deep Convolutional GAN (DCGAN). The paper addresses the challenge of dynamic and time-varying network intrusion data, which often leads to a lack of training instances and high false detection rates. To mitigate this, the technique utilizes the Difficult Set Sampling Technique (DSSTE) to reduce noise samples in the majority category and DCGANs to boost the minority sample size.

3.3. Data Augmentation

Data Augmentation expands training datasets by transforming existing samples or generating synthetic data. This approach increases dataset diversity, which enhances model generalization and robustness to input variations. While some studies apply Data Augmentation to address class imbalance, this section highlights its broader applications.

The study by [51] explores the use of GANs to synthesize cyber intrusion alerts. This technique captures intricate feature dependencies and rare attack behaviors without requiring prior knowledge, thereby improving cyber threat analysis. Specifically, the research evaluates the effectiveness of Wasserstein-GAN with Gradient Penalty and Mutual Information in generating realistic alerts.

Similarly, Liu et al. [56] address the scarcity of cyber threat data in the space domain by generating synthetic threat data to support intrusion detection and protection. Their approach extends existing data generation methods, including GANs and variational autoencoders, to create data tailored for space systems. Le et al. [57] introduce IMIDS, an intelligent IDS for IoT devices. IMIDS leverages a CNN and a CGAN to mitigate the shortage of training data. Experimental results show IMIDS achieves high detection rates across nine types of cyberattacks, outperforming competing methods and demonstrating its effectiveness in enhancing IoT security.

Table 2
Summary of offensive strategies using generative AI.

Offensive strategy	Description	Attack target	Literature findings
Input Perturbation	Subtle changes to input data designed to mislead machine learning models, often using GANs to generate adversarial examples.	IDS systems, Malware detectors, DNNs	[64], [65], [66], [67], [68], [69], [70]
Data Poisoning	Injecting malicious data into training datasets to degrade the performance of machine learning models.	IoT networks, Federated learning	[71], [72]
Spoofing	Generating fake inputs that mimic legitimate data to deceive machine learning models, often in wireless communication systems.	Wireless signals, Biometric systems	[73],[74]

Mimura [52] proposes a method to improve the sensitivity of minority class detection in macro malware analysis. While this work also addresses class imbalance, it focuses primarily on the challenges posed by small datasets in malware detection. The study emphasizes the importance of continuously collecting comprehensive data samples to build more accurate models. Kang et al. [54] tackle a similar challenge by proposing a DNN-based malware classifier that employs GANs for Data Augmentation, thereby enhancing training efficacy.

GANs have also been utilized for Data Poisoning defense. In [53], a GAN-based solution is proposed to address the limitations of existing defense techniques, which are often tailored to specific types of attacks. By augmenting training datasets with synthetic data, this approach provides a more generalized, attack-agnostic solution against Data Poisoning.

Liu et al. [55] propose SFE-GACN, a method for detecting unknown attacks in network traffic intrusion scenarios. This technique leverages SFE to summarize the context of network traffic sessions. Additionally, GACN supervises sample generation to maintain intracategory diversity, resulting in improved detection performance compared to state-of-the-art methods.

Alabsi et al. [58] present an IDS based on CTGAN to detect DDoS and DoS attacks in IoT networks. By generating synthetic traffic resembling legitimate patterns, this CGAN-based IDS improves detection capabilities. Similarly, Alabdulwahab et al. [59] employ CTGAN to create synthetic datasets for machine learning-based IDS in IoT environments, addressing the challenge of acquiring large-scale, real-world datasets.

3.4. Data Obfuscation

Data Obfuscation techniques aim to conceal or modify sensitive information that could otherwise be exploited by attackers. Within the current body of research, relatively few studies focus specifically on Generative AI for data obfuscation. Fathi-Kazerooni and Rojas-Cessa [60] propose “GAN tunnel”, a traffic masking method that protects application identities from adversarial Internet traffic classifiers (ITCs). Such systems might be exploited by malicious actors for website fingerprinting and detection of network protocols, inferring users’ information. GAN tunnel generates fake traffic and encapsulates actual user traffic to prevent classification from adversarial ITCs. Romanelli et al. [61] focus on obfuscating sensitive location information while preserving its utility. They propose a GAN-based approach that includes a generator to produce an optimal obfuscation mechanism and a classifier that attempts to de-obfuscate the data. By letting the two networks compete until equilibrium is reached, the mechanism continuously improves its protective capabilities.

Lastly, Desai et al. [62] introduce a privacy-preserving architecture for fine-grained power consumption data in smart grids. They employ a GAN and an obfuscator to generate synthetic time series data that replace existing appliance signatures while minimizing energy discrepancies, effectively mitigating privacy breaches. Their approach ensures that the synthetic data closely resembles real-world data while being less complex, making it suitable for IoT environments and smart city applications.

4. Offensive strategies using Generative AI

This section examines studies that highlight the potential malicious applications of Generative AI in security contexts. The identified categories are elaborated in the subsections below and summarized in Table 2.

4.1. Input Perturbation

In ML, Input Perturbation involves small, carefully crafted modifications to input data to cause misclassification or degrade a model’s performance. These alterations, often imperceptible to humans, can significantly disrupt a model’s output and make it vulnerable to adversarial attacks.

The disruptive potential of Input Perturbation is well-documented. Liu et al. [64] propose a two-stage GAN (TSGAN) with semantic content constraints to generate high-quality adversarial examples that achieve high attack success rates. Alshahrani et al. [67] explore adversarial attacks on supervised ML-based network intrusion detection systems, demonstrating a notable decline in detection accuracy caused by Input Perturbation. Similarly, Aldhaheri and Alhuzali [69] introduce SGAN-IDS, a framework that uses GANs and self-attention mechanisms to craft adversarial attack flows that evade detection and significantly lower the effectiveness of various IDSs.

Chauhan et al. [65] leverage GAN-based Input Perturbation to alter the attack profiles of DDoS samples, rendering them undetectable to ML-based IDSs. However, this approach does not address whether the altered samples retain their functionality. In contrast, Zhao et al. [66] present AttackGAN, an adversarial attack model designed to target black-box IDSs effectively while ensuring that the modified network traffic maintains its intended functionality. This work underscores the importance of preserving the operational integrity of attacks during perturbation.

In the context of malware, [68] proposes IF-MalEvade, a framework combining GANs and deep reinforcement learning to generate adversarial malware samples with subtle modifications. These samples evade detection by black-box ML-based malware detectors while retaining their malicious behavior. This approach highlights the effectiveness of modifying key features of malware samples to bypass detection systems without compromising functionality.

For physical adversarial attacks, Ma et al. [70] introduce GAN-VS, a framework tailored for the AIoT domain. GAN-VS generates adversarial patches strategically placed in critical regions of images, reducing their visibility while increasing their effectiveness in deceiving DNNs. This approach balances camouflage and impact, making it particularly effective for physical attacks.

4.2. Data Poisoning

Data Poisoning attacks involve injecting malicious data into a training dataset to undermine a model’s performance or integrity. This can be achieved by altering existing training samples or influencing the data collection process to bias the model toward specific, incorrect predictions.

Liu et al. [71] address poisoning attacks targeting IDS with AIGAN, a GAN designed to generate adversarial data for anomaly attacks on IoT networks. The effectiveness of AIGAN's generator is evaluated using the Jaccard similarity index to compare real and synthetic data, as well as F1-scores across various ML algorithms. Their findings emphasize the vulnerability of NIDS when exposed to poisoned data, revealing significant susceptibility to adversarial perturbations.

In another study, Khowaja et al. [72] explore vulnerabilities in federated learning for vehicular networks, a critical component for advancing beyond 5G and 6G technologies. They propose SPIN, a simulated poisoning and inversion network, which reconstructs data intercepted by roadside units (RSUs) and trains a GAN to enhance data generation. SPIN demonstrates a notable reduction in model accuracy on publicly available datasets, showcasing its effectiveness and the potential real-world implications of poisoning attacks in federated learning. This research highlights the urgent need for robust defenses against model poisoning in critical domains such as vehicular networks.

4.3. Spoofing

Spoofing attacks involve creating fake inputs that mimic legitimate data to deceive a model. These attacks may include generating synthetic data or crafting inputs that exploit vulnerabilities in a model's decision-making process. Spoofing is particularly concerning in security-sensitive applications, such as biometric authentication systems and fraud detection algorithms.

Shi et al. [73] propose a deep learning-based Spoofing attack targeting physical-layer signal authentication in wireless communication systems. Their approach models the adversary as a transmitter–receiver pair and employs a GAN to generate synthetic wireless signals indistinguishable from legitimate transmissions. By jointly modeling waveform, channel, and radio hardware effects, the attack significantly increases the probability of misclassifying Spoofing signals as authentic, demonstrating its effectiveness in bypassing physical-layer signal authentication.

Similarly, Zerhouni et al. [74] highlight vulnerabilities in wireless communication systems by using GANs to create counterfeit wireless signals that mimic legitimate ones. These signals are misclassified as authentic by CNN-based receivers. The study applies GANs to model filtered multi-carrier waveforms, such as orthogonal frequency-division multiplexing (OFDM) and filtered orthogonal FDM (F-OFDM). Their successful evasion attack demonstrates how GAN-generated signals can deceive target receivers, emphasizing the critical need for preventive measures to counter GAN-based Spoofing attacks in wireless communication systems.

5. Gap analysis of Generative AI in cybersecurity

In the previous section, we analyzed the defensive and offensive capabilities of Generative AI identified through the SLR. Although much of the research emphasizes using Generative AI for defensive strategies in IDS, a significant gap remains in understanding its offensive applications. Most studies focus on Input Perturbation, with comparatively less attention given to Data Poisoning and Spoofing attacks.

In this paper, we propose a novel offensive strategy: Attack Obfuscation. This approach introduces a method to generate synthetic traffic that is merged with legitimate samples, which remain unaltered. The primary objective is to deceive NN-based IDSs by complicating the detection of actual attacks through the integration of adversarial synthetic data. The key differences from existing methods are detailed below.

- Comparison with Input Perturbation: Input Perturbation involves modifying legitimate or malicious samples with small, often imperceptible changes to evade detection. In contrast, Attack Obfuscation leaves both legitimate and malicious samples unchanged. Instead, it generates synthetic traffic that blends with existing network activity, making it difficult for IDS classifiers to distinguish between benign, malicious, and synthetic traffic.
- Comparison with Data Poisoning:Data Poisoning manipulates training datasets to degrade IDS performance during inference, often by introducing mislabeled or adversarial samples during training. Attack Obfuscation, however, exclusively targets the test phase, leaving the training data unaltered. By obfuscating attack signatures at the testing stage, the method exploits an IDS's dependency on clear patterns for effective detection.
- Comparison with Spoofing: Spoofing generates noise or adversarial samples designed to mimic legitimate traffic and evade detection by being misclassified as benign. Unlike Spoofing, Attack Obfuscation does not focus on creating instances that are directly misclassified as legitimate. Instead, it seeks to introduce controlled adversarial traffic that masks the actual malicious activity, reducing the IDS's overall detection performance.

Leveraging CGANs, the proposed method generates synthetic traffic that adheres to specific conditions. This ensures a realistic blend with existing network traffic and provides fine-grained control over the generated data. Unlike Input Perturbation or Data Poisoning, Attack Obfuscation preserves the integrity of original traffic (both benign and malicious) and focuses on obfuscating detection mechanisms through augmentation. The goal is not just evasion but the deliberate obfuscation of attack patterns, making it challenging for NN-based IDSs to distinguish real attacks from synthetic noise.

The experiments detailed in Section 6 aim to address the gap identified in the state-of-the-art. By focusing on the offensive capabilities of Generative AI for concealing attacks, the proposed methodology evaluates the effectiveness of CGANs in generating realistic and deceptive network traffic to evade detection by NN-based IDSs. Two distinct scenarios (training and attack generation on the same dataset, and on separate datasets, respectively) are explored to assess the robustness and versatility of the Attack Obfuscation technique. CGANs offer a significant advantage over traditional GANs by allowing the controlled generation of synthetic data based on specified conditions. This enables the creation of realistic and contextually relevant traffic patterns, which are crucial for effective obfuscation. This methodology bridges the existing research gap and highlights the potential risks associated with the misuse of Generative AI in offensive cybersecurity strategies.

6. Proposed CGAN-based attack obfuscation

Following the state-of-the-art analysis, this section shows the proposed experimental approach. The study aims to evaluate the effectiveness of Generative AI in both enhancing IDS and obfuscating attacks to deceive such systems. In comparison with the existing literature, the proposed research presents a novel attack obfuscating approach targeting ML-based IDS, highlighting an unexplored area and providing insights for further defensive research. The workflow of the experiments is schematized in Fig. 7.

The experiment involves two main components: a NN-based IDS and a CGAN-based obfuscator. This setup simulates real-world scenarios where attackers use fake traffic to bypass IDS detection. The procedure follows multiple steps to achieve effective obfuscation.

- **Training and Testing the NN-based IDS:** The IDS was trained using a dataset D_{train}, containing both benign and malicious network traffic instances with associated labels. The IDS is trained to classify incoming network traffic as either benign or malicious based on learned patterns. Subsequently, the performance of the trained IDS is evaluated using a separate labeled dataset D_{test}, assessing metrics such as precision, recall, F1-score, and overall accuracy.
- **Training the CGAN-based Obfuscator:** The Obfuscator is trained to generate synthetic network traffic samples that mimic legitimate traffic patterns observed in the training data.

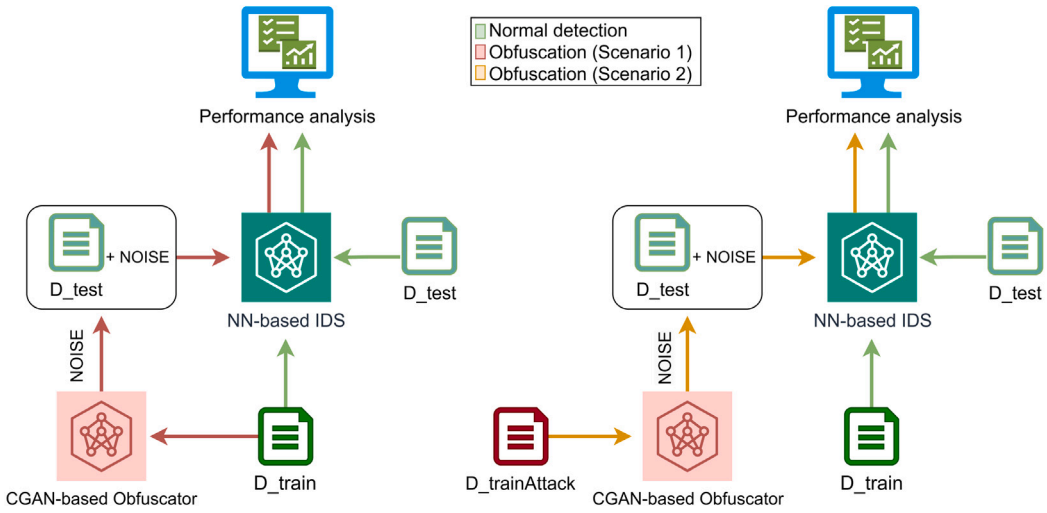


Fig. 7. Workflow of the proposed approach using a NN-based IDS and a CGAN-based Obfuscator. In Scenario 1 (left), the attacker uses the same training data as the IDS (D_{train}) to generate obfuscated attack samples. In Scenario 2 (right), the attacker uses a separate training dataset ($D_{trainAttack}$) with similar features to obfuscate attacks. The green arrows indicate normal IDS training and testing, the red arrows show obfuscation in Scenario 1, and the orange arrows depict obfuscation in Scenario 2. The CGAN-based Obfuscator generates fake traffic to bypass the IDS without altering the actual attacks.

- **Generating Fake Traffic for Attack Obfuscation:** Once trained, the Obfuscator is employed to generate fake traffic samples, which are appended to D_{test} to create a new test dataset, $D_{testObfuscated}$, which was then used to evaluate the performance of the IDS in the presence of obfuscated attacks.
- **Testing the NN-based IDS against obfuscated traffic:** Two different scenarios are considered in this phase. First, the IDS and the obfuscator are trained using the same dataset D_{train} (Scenario 1). Subsequently, assuming that the attacker does not have access to D_{train}, a separate dataset $D_{trainAttack}$ with similar features and labels was employed to train the Obfuscator (Scenario 2).

By comparing the performance of the IDS in both scenarios, the effectiveness of the CGAN-based obfuscator in successfully obfuscating attacks and causing performance degradation in the IDS was demonstrated. In the following, the main components of the proposed approach are detailed.

6.1. Dataset selection

Within the experiments, three different datasets were employed to replicate the methodology against various kinds of traffic and attacks. The individual datasets were split in D_{train}, D_{test}, and $D_{trainAttack}$, with D_{train} being balanced to ensure the best training for the IDS. First, the CIC IoT dataset 2023 [75] (D_1) was selected due to its novelty and completeness. The original dataset comprehends 33 attacks targeting 105 IoT devices. However, for the scope of the present research, only a subset of attacks have been extracted and used for the actual experiments, including MITM, and three variants of Mirai. Specifically, Mirai greeth flood, Mirai greip flood, and Mirai uddplain. The second dataset employed was the LUFlow Network Intrusion Detection Data Set [76] (D_2), which contains emerging attack vectors through the composition of honeypots within Lancaster University's address space. This dataset was selected to address the effectiveness of the proposed approach against novel attack patterns. Finally, to assess performances against SQL Injection attacks, the SQL Injection Attack Netflow [77] dataset (D_3) was employed as well.

6.2. NN-based IDS: Architecture and implementation

One of the key components of the proposed approach is a ML-based classifier, leveraging deep learning to discriminate between benign and malicious traffic. The implementation uses the Python library PyTorch for model construction and training. Unlike traditional IDSs that rely on static rule sets or signature-based detection methods, NN-based approaches can autonomously learn and adapt to dynamic patterns of network traffic. At its core, the IDS comprises a NN architecture designed to analyze network traffic data and classify it into distinct categories, in accordance with the training dataset. The architecture consists of interconnected layers of artificial neurons, each equipped with activation functions that enable the extraction of meaningful features from input data. Specifically, rectified linear units (ReLU) are employed as activation functions in the hidden layers to introduce non-linearity, while a softmax activation function is applied in the output layer to produce class probabilities. Throughout the development process, data pre-processing, feature engineering, and model optimization were employed to ensure the robustness and reliability of the IDS. Pre-processing steps included handling missing values, eliminating duplicate entries, and performing dimensionality reduction to enhance the efficiency of the model. Following data preparation, the NN model was trained using a combination of labeled training data and supervised learning techniques. The training process involved multiple epochs, with the number of epochs carefully chosen to balance model convergence and computational efficiency. Additionally, the model was optimized using the Adam optimizer with a learning rate of 0.01, which facilitates efficient parameter updates and convergence. During training, the model learned to distinguish between benign and malicious network traffic patterns by iteratively adjusting its internal parameters to minimize classification errors. A detailed summary of the implementation is shown in Table 3.

6.3. CGAN-based obfuscator: Architecture and implementation

To test the offensive strategy, the CGAN-based Obfuscator was designed and implemented to generate synthetic network traffic instances, meant to conceal the presence of actual attacks within the network data. The module includes two components: a Generator and a Discriminator, implemented using the Python library Keras. The heart of the Obfuscator lies in its Generator Network. This NN is in charge of synthesizing traffic instances based on random noise vectors (latent space) and conditional labels representing different classes of network traffic. It features a multi-layered architecture consisting of densely connected layers with ReLU activation functions, enabling the introduction of non-linearity. The final sigmoid activation layer facilitates the generation of binary outputs, effectively emulating the characteristics

Table 3
Implementation details for the NN-based IDS architecture, training, and testing.

Category	Details
Neural Network Architecture	- **Input Layer**: 15 features - **Hidden Layer**: 1 hidden layer with 8 neurons, ReLU activation - **Output Layer**: Number of neurons = number of output classes, dynamically defined using LabelEncoder) - **Activation Functions**: ReLU for hidden layer, no explicit activation for output (handled by CrossEntropyLoss)
Loss Function	- Cross-entropy loss
Optimizer	- Adam optimizer, learning rate = 0.01
Training Process	- **Epochs**: 100 epochs, with loss printed every 10 epochs - **Batch Size**: entire dataset in each iteration - **Gradient Descent**: Performed after each epoch using Adam optimizer
Data Preprocessing	- **Label Encoding**: Applied using LabelEncoder for converting categorical labels to numerical form - **Dimensionality Reduction**: SelectKBest used with ANOVA to select 15 features - **Feature Scaling**: Standardized using StandardScaler for both training and test sets
Evaluation Process	- **Metrics**: Precision, Recall, F1-score using classification report from Sklearn - **Test Data**: CIC IoT dataset 2023, LUFlow Network Intrusion Detection Data Set, SQL Injection Attack Netflow - **Evaluation Strategy**: Predictions made using softmax, evaluated with ground truth labels from test set

of legitimate network traffic. On the other hand, the Discriminator Network is crucial in discerning between real (legitimate) and fake (synthetic) traffic instances. Implemented as a binary classifier with multiple densely connected layers and a sigmoid activation function in the output layer, the discriminator predicts the probability of a traffic instance being real or fake. During training, the Generator attempts to deceive the Discriminator by producing synthetic traffic instances that closely resemble real ones, while the Discriminator tries to accurately classify between real and fake traffic. This adversarial interplay between the Generator and Discriminator allows the Generator to produce realistic samples, which are subsequently appended to the test data, without altering the existing samples.

6.4. Results

This section summarizes the experimental results, evaluated based on Precision (Eq. (1)), Recall (Eq. (2)), F1-Score (Eq. (3)), Accuracy (Eq. (4)), Balanced Accuracy (BA) (5), Matthews Correlation Coefficient (MCC) (6), and Geometric Mean (G-mean) (7). In the following equations, True Positive (TP), True Negative (TN), False Positive (FP) and False Negative (FN) are denoted using the acronyms.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{1}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{2}$$

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3}$$

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \tag{4}$$

$$\text{BA} = \frac{\text{Sensitivity} + \text{Specificity}}{2} \tag{5}$$

$$\text{MCC} = \frac{\text{TP} \times \text{TN} - \text{FP} \times \text{FN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FP})}} \tag{6}$$

$$\text{G-mean} = \sqrt{\text{Sensitivity} \times \text{Specificity}} \tag{7}$$

Tables 4, 5, and 6 showcase the performance of the IDS against D_{test}, and the $D_{testObfuscated}$ generated using D_{train} (Scenario 1) and $D_{trainAttack}$ (Scenario 2), respectively, for the three different datasets employed. For the sake of brevity, shorter names are used to denote some of the classes and metrics in the tables: MITM, Mirai greeth flood (Mirai-GF), Mirai greip flood (Mirai-GP), Mirai uddplain (Mirai-UDP), and SQL Injection (SQL-Inj). For the metrics: Precision (Prec), Recall (Rec), Accuracy (Acc). The results are commented on in the next Section.

6.5. Statistical validation of IDS performance

To ensure that the performance of the NN-IDS is not overly sensitive to the random initialization of weights, statistical validation of the model's behavior was performed across multiple runs. Specifically, the model was trained on D_1 using the same configuration but with different random seeds for initializing the network weights. This process was repeated 10 times to account for variability in the model's initial conditions. During each run, the model was trained for 100 epochs, and two key performance metrics were recorded (Accuracy and F1-score). The NN was evaluated on the clean test set after each run, and the results for both metrics were collected. To assess the statistical significance of any variability in performance across the different runs, we applied one-way ANOVA. ANOVA is used to test whether the means of several groups are statistically different from each other. In this context, we used it to compare the performance metrics (Accuracy and F1-score) across the 10 different runs, with the null hypothesis being that there is no significant difference in the performance across runs due to random weight initialization.

The p-values obtained from the ANOVA test for both Accuracy and F1-score were as follows:

- p-value for Accuracy: 0.9395058502783618
- p-value for F1-score: 0.8598826693254419

Since both p-values were greater than the significance threshold of 0.05, we conclude that there is no statistically significant difference in the performance of the model across the different runs. This indicates that the model's performance is stable and consistent across various initial conditions, and the random initialization of weights does not significantly affect the results.

6.6. Comparison with random forest-based IDS

To further validate the effectiveness of the CGAN-based obfuscator, the same experimental approach was replicated using a RF classifier as the IDS, instead of the NN-based model. This additional comparison aimed to assess the robustness of different IDS architectures when subjected to obfuscated attacks. For consistency, the experiments were conducted using the same subset of attacks from the CIC IoT dataset 2023 (D_1). The Random Forest classifier was implemented using the Python library Scikit-learn. The model consists of multiple decision trees, each trained on different subsets of the training data and features, to create a robust and generalized classifier. The majority voting mechanism across the trees determines the final classification output, making it less susceptible to overfitting and noise in the data. The performance metrics of the RF-based IDS before and after obfuscation are presented in Table 7. Similar to the NN-based approach, the results are evaluated in the context of normal detection, Scenario 1, and Scenario 2.

Table 4
Summary of IDS performance before and after the obfuscation using the IoT23 Dataset D_1.

	Normal detection							Scenario 1							Scenario 2						
	Prec	Rec	F1-score	Acc	BA	MCC	G-mean	Prec	Rec	F1-score	Acc	BA	MCC	G-mean	Prec	Rec	F1-score	Acc	BA	MCC	G-mean
Benign	0.83	0.99	0.9					0.81	0.68	0.74					0.85	0.68	0.75				
MITM	0.87	0.33	0.48					0.84	0.08	0.15					0.89	0.16	0.27				
Mirai-GF	0.58	0.93	0.71	**0.76**	0.99	0.74	0.99	0.58	0.65	0.61	**0.54**	0.68	0.36	0.68	0.58	0.61	0.6	**0.55**	0.68	0.42	0.68
Mirai-GP	0.55	0.11	0.18					0.63	0.03	0.06					0.55	0.08	0.14				
Mirai-UDP	0.99	0.99	0.99					0.41	0.99	0.58					0.41	0.99	0.58				

Table 5
Summary of IDS performance before and after the obfuscation using the LuFlow Dataset D_2.

	Normal Detection							Scenario 1							Scenario 2						
	Prec	Rec	F1-score	Acc	BA	MCC	G-mean	Prec	Rec	F1-score	Acc	BA	MCC	G-mean	Prec	Rec	F1-score	Acc	BA	MCC	G-mean
benign	0.99	0.83	0.9					0.99	0.01	0.02					0.97	0.04	0.08				
malicious	0.72	0.99	0.84	**0.82**	0.62	0.82	0.62	0.39	0.99	0.56	**0.39**	0.34	0.14	0.34	0.55	0.91	0.68	**0.45**	0.34	0.21	0.34
outlier	0.83	0.03	0.05					0.09	0.01	0.01					0.2	0.07	0.1				

Table 6
Summary of IDS performance before and after the obfuscation using the SQL Injection Dataset D_3.

	Normal Detection							Scenario 1							Scenario 2						
	Prec	Rec	F1-score	Acc	BA	MCC	G-mean	Prec	Rec	F1-score	Acc	BA	MCC	G-mean	Prec	Rec	F1-score	Acc	BA	MCC	G-mean
Benign	0.99	0.41	0.58	**0.71**	0.7	0.7	0.63	0.5	0.91	0.65	**0.5**	0.5	∼0	0.28	0.5	0.78	0.61	**0.5**	0.5	∼0	0.41
SQL-Inj	0.63	0.99	0.77					0.5	0.09	0.15					0.5	0.22	0.31				

Table 7
Summary of RF-based IDS performance before and after obfuscation using the IoT23 Dataset D_1.

	Normal Detection							Scenario 1							Scenario 2						
	Prec	Rec	F1-score	Acc	BA	MCC	G-mean	Prec	Rec	F1-score	Acc	BA	MCC	G-mean	Prec	Rec	F1-score	Acc	BA	MCC	G-mean
Benign	0.92	0.99	0.95					0.95	0.15	0.23					0.93	0.2	0.33				
MITM	0.92	0.7	0.8					0.23	0.74	0.35					0.23	0.74	0.35				
Mirai-GF	0.98	0.9	0.94	**0.94**	0.85	0.98	0.83	0.4	0.04	0.07	**0.45**	0.35	0.36	0.33	0.4	0.04	0.07	**0.47**	0.47	0.28	0.38
Mirai-GP	0.88	0.97	0.92					0.2	0.6	0.3					0.21	0.6	0.31				
Mirai-UDP	0.99	0.99	0.99					0.6	0.3	0.4					0.6	0.3	0.4				

7. Discussion

The findings in the literature underscore the growing importance of Generative AI techniques in improving cybersecurity defenses and the escalating arms race between defenders and adversaries. The effectiveness of these techniques in enhancing the resilience of IDSs against various cyber threats has been well-documented across different domains, including network security, malware detection, and anomaly detection. On the other hand, the literature also highlights the emerging threat posed by offensive strategies leveraging Generative AI, such as Input Perturbation, Data Poisoning, and Spoofing attacks. These malicious techniques exploit vulnerabilities in IDSs and other cybersecurity mechanisms, emphasizing the need for continuous innovation in defensive strategies to counter evolving threats effectively. The experimental results provide insights into the performance of a NN-based IDS under different scenarios: normal detection, and two varieties of obfuscated attacks. In normal detection scenarios, the IDS demonstrates good performance, identifying known patterns of malicious activity and effectively differentiating them from legitimate network traffic. In Scenario 1, where the attacker can use D_{train}, the performance of the IDS shows a significant decrease in detection accuracy compared to normal detection for the entire selection of datasets employed. In particular, a decrease in precision compared to normal detection is observed across the datasets for most of the classes, with the exception of D_1, where only Mirai-UDP seems affected. Similarly, recall decreases. These parameters indicate an increase in false positive and false negative classifications. The F1-score, being the harmonic mean of precision and recall, also decreases compared to normal detection. This decrease reflects the overall decline in the system's ability to accurately detect both malicious and benign instances. The overall accuracy of the IDS drops in Scenario 1, highlighting the negative impact of the obfuscation on the system's performance, where the best results are practically equal to random guessing. In Scenario 2, where the attacker cannot access D_{train} and uses $D_{trainAttack}$, the attacks are still successfully concealed, despite a slightly better performance of the IDS in comparison to Scenario 1, as expected. Notably, the experimental results reveal the vulnerability of NN-based IDSs to obfuscated attacks crafted using Generative AI: despite the IDS's initial robustness in normal detection scenarios, the effectiveness of the system is significantly compromised when faced with adversarially crafted inputs, with even unaltered attack samples being wrongly classified.

The statistical validation using one-way ANOVA demonstrates that the variability in performance metrics across multiple runs is not significant. Therefore, the NN's performance is robust against random initialization of weights, and the reported results are reliable and reproducible.

The RF-based IDS demonstrated higher baseline performance compared to the NN-based IDS across most metrics, indicating its robustness in detecting various attack types in D_1. However, the impact of obfuscation by the CGAN-based obfuscator was still significant, particularly in Scenario 1, where the precision, recall, and overall accuracy experienced notable degradation. Scenario 2 showed a similar trend, with the IDS performance further impacted by the different training datasets used by the obfuscator.

The findings indicate that with the incorporation of Generative AI into cybersecurity, both offensive and defensive actors find themselves on equal footing. This convergence effectively returns the landscape to its initial state, where traditional rule-based IDSs emerge as primary defense mechanisms. However, this realization underscores the need for continued exploration and innovation within the cybersecurity domain to effectively counter the ever-evolving threats.

7.1. Possible defensive countermeasures

Generative AI, particularly techniques like CGANs, has shown considerable potential for offensive cybersecurity strategies, including attack obfuscation, which makes detecting malicious activities more difficult for traditional IDS. One potential defensive measure is the use of GAN-based adversarial training. By incorporating adversarial samples during IDS training, the system can learn to recognize not only typical attack patterns but also those generated by obfuscation techniques. This would involve training the IDS with both obfuscated and non-obfuscated attacks, improving its ability to detect generative adversarial attacks. Another approach is anomaly detection using advanced unsupervised learning algorithms that can identify deviations from normal network behavior without relying on predefined attack signatures. Furthermore, behavioral monitoring systems that can track and analyze long-term network trends could help in identifying anomalies that evade short-term detection mechanisms. By continuously profiling the behavior of both legitimate and malicious traffic, it might be possible to detect subtle irregularities introduced by attack obfuscation. Another promising defense strategy is meta-learning (learning to learn), where the IDS can adapt and respond dynamically to newly evolving threats, including those generated by AI. This strategy could provide a more agile defense by continuously learning and adjusting to new obfuscation patterns as they emerge in real-time attacks. Future research should also focus on defensive GANs, which could be used to counter offensive GANs by generating adversarial traffic for defensive purposes. Such GANs could simulate both offensive and defensive scenarios, helping to predict and mitigate the impact of new obfuscation strategies.

7.2. Threat to validity and future work

While the experiments show promising results, it is important to acknowledge limitations and threats to validity. Firstly, the experiments were conducted in a controlled environment with simulated attack scenarios, which may not fully capture the complexity and variability of real-world cyber threats. Additionally, the performance of the IDS may vary depending on factors such as the specific dataset used for training and testing, the configuration of the CGAN, and the nature of the attacks employed. Furthermore, the evaluation metrics used to assess the performance of the IDS, such as detection accuracy and false positive rate, provide only a partial understanding of its effectiveness in real-world scenarios. Other factors, such as detection latency, scalability, and resilience to zero-day attacks, could require more investigation to comprehensively evaluate the capabilities of the IDS, leading to possible further research. Additional experiments could be conducted using even more diverse datasets and attack scenarios to evaluate the robustness of the IDS across different environments and threat landscapes. Moreover, novel defensive strategies, such as ensemble learning, deep reinforcement learning, and explainable AI (xAI), could be explored to enhance the resilience of IDSs against adversarial manipulation. Research efforts could focus on developing more sophisticated Generative AI models capable of generating highly realistic and deceptive adversarial samples, thereby posing greater challenges to IDSs. Additionally, investigating the feasibility of incorporating real-time feedback mechanisms and adaptive learning algorithms into IDSs could further improve their ability to adapt to evolving threats and mitigate the impact of adversarial attacks.

Another significant consideration is the computational complexity of the proposed approach, particularly regarding the training of CGANs and the real-time generation of synthetic traffic. The training phase of CGANs is computationally demanding, as it involves iterative adversarial optimization between the generator and the discriminator. This process can be resource-intensive, often requiring high-performance GPUs or TPUs to manage the high-dimensional data involved in network traffic simulations. Furthermore, hyperparameter tuning to ensure convergence and performance stability adds a layer of computational overhead. While generating synthetic traffic using a pre-trained CGAN is considerably faster than training, the inference process can still introduce latency, particularly in high-throughput environments where large volumes of network traffic are analyzed in real-time.

Addressing these computational challenges is essential for translating the proposed approach into real-world scenarios. Future research could explore strategies to mitigate these demands, such as employing transfer learning to fine-tune pre-trained CGANs for specific attack scenarios, optimizing CGAN architectures to reduce their computational footprint, or leveraging distributed systems to accelerate both training and inference processes. Techniques like model compression or quantization could also be investigated to facilitate real-time deployment by reducing the complexity of CGAN inference.

Translating the proposed approach to real-world environments introduces additional complexities. Real-world networks are characterized by a high degree of traffic variability, integration with legacy systems, and the constant emergence of new, sophisticated threats. As such, real-time deployment would need to balance computational efficiency with scalability, especially when processing large volumes of data. The complexity of CGAN-based models, while powerful, could present challenges in terms of resource consumption and real-time responsiveness.

To operate effectively in real-world conditions, the system would require robust feedback mechanisms and adaptive learning algorithms capable of adjusting to new attack patterns autonomously. Future work should evaluate the method's performance in live environments, exploring key dimensions such as real-time adaptability, computational costs, and resilience to zero-day attacks. Additionally, collaborative testing with industry partners would provide insights into how the system handles diverse and evolving threat landscapes.

As Generative AI continues to mature, both attackers and defenders are likely to leverage these technologies in an ongoing arms race. Offensive strategies, such as attack obfuscation through CGANs, demonstrate the ability of AI to deceive IDS systems by generating synthetic traffic that mimics legitimate behavior. However, defenders can also exploit generative AI to create more robust IDS models capable of detecting increasingly sophisticated attacks.

Adaptive learning models could be crucial in this scenario. Just as attackers use AI to dynamically generate new attack vectors, defenders must adopt equally flexible and adaptive systems. This means that static detection models, reliant on predefined rules or attack signatures, may become obsolete in the near future. Instead, continuous learning systems that can autonomously adapt to new types of attacks will likely be the foundation of next-generation cybersecurity defenses.

Moreover, AI-driven cyberattacks raise important questions about ethics and regulation. As generative AI becomes more accessible, it could potentially democratize the ability to launch sophisticated attacks, lowering the barrier to entry for malicious actors. Therefore, the cybersecurity community, alongside policymakers, must consider how to balance innovation with control, ensuring that the benefits of AI are not overshadowed by the risks posed by its misuse.

Finally, the study underscores the need for collaboration between the research community, industry, and regulatory bodies to develop frameworks and guidelines for the responsible use of AI in cybersecurity. Defensive AI must not only be reactive but also proactive, anticipating the next generation of AI-based threats and developing preemptive countermeasures.

8. Conclusion

In cybersecurity, the advent of Generative AI has brought a paradigm shift in both defense and offense. This study examined its applications in security, analyzing both defensive strategies that strengthen intrusion and anomaly detection, and offensive tactics that seek to undermine them. The literature review revealed a set of defensive applications of Generative AI, with a primary focus on Adversarial

Training, Dataset Balancing, Data Augmentation, and Data Obfuscation. Adversarial Training emerged as a potent technique to enhance IDS robustness by exposing models to adversarial samples during training, improving their ability to generalize and resist attacks. Additionally, Dataset Balancing and Data Augmentation techniques addressed challenges such as class imbalance and data scarcity, improving the performance of IDSs across various domains. Furthermore, Data Obfuscation techniques leveraged Generative AI to conceal sensitive information and protect against malicious exploitation. On the offensive side, Generative AI was shown to be a valid tool for crafting attacks capable of evading IDSs. Input Perturbation, Data Poisoning, and Spoofing emerged as prominent strategies employed by attackers to subvert IDSs and compromise network security. These offensive techniques underscored the pressing need for robust defense mechanisms capable of mitigating the risks posed by adversarial attacks. While the SLR showed increasing interest in the topic and a good variety of approaches, there is still a gap that needs to be addressed. For instance, attackers could use GANs and CGANs to obfuscate malicious data and deceive IDSs. This was the focus of the experimental approach presented in this work. Unlike existing state-of-the-art approaches that primarily focus on altering attack signatures, poisoning training datasets, or generating spoofed samples that are classified as legitimate, our experimental approach focuses on using CGANs to generate synthetic network traffic instances that closely resemble legitimate traffic, camouflaging malicious activities within the network and fooling the NN-based classifier. Through a series of experiments conducted across multiple datasets, the effectiveness of the proposed approach was demonstrated in obfuscating attacks and causing performance degradation in IDSs in two variations of the same strategy. These results indicate the need for further research and development of security strategies, that must take into consideration the fast evolution of AI techniques for offensive purposes.

CRediT authorship contribution statement

Luigi Coppolino: Supervision, Conceptualization. **Salvatore D'Antonio:** Writing – review & editing. **Giovanni Mazzeo:** Writing – review & editing. **Federica Uccello:** Writing – original draft, Validation, Data curation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Luigi Coppolino reports financial support was provided by European Union. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

[1] J. Halvorsen, C. Izurieta, H. Cai, A.H. Gebremedhin, Applying generative machine learning to intrusion detection: A systematic mapping study and review, ACM Comput. Surv. (2024).
[2] A. Arora, Shantanu, A review on application of GANs in cybersecurity domain, IETE Tech. Rev. 39 (2) (2022) 433–441.
[3] A. Dunmore, J. Jang-Jaccard, F. Sabrina, J. Kwak, A comprehensive survey of generative adversarial networks (gans) in cybersecurity intrusion detection, IEEE Access (2023).
[4] X. Xia, X. Pan, N. Li, X. He, L. Ma, X. Zhang, N. Ding, GAN-based anomaly detection: A review, Neurocomputing 493 (2022) 497–535, http://dx.doi.org/10.1016/j.neucom.2021.12.093, URL https://www.sciencedirect.com/science/article/pii/S0925231221019482.
[5] C. Zhang, S. Yu, Z. Tian, J.J. Yu, Generative adversarial networks: A survey on attack and defense perspective, ACM Comput. Surv. 56 (4) (2023) 1–35.
[6] S. Wang, R.K. Ko, G. Bai, N. Dong, T. Choi, Y. Zhang, Evasion attack and defense on machine learning models in cyber-physical systems: A survey, IEEE Commun. Surv. Tutor. (2023).
[7] K. Petersen, S. Vakkalanka, L. Kuzniarz, Guidelines for conducting systematic mapping studies in software engineering: An update, Inf. Softw. Technol. 64 (2015) 1–18.
[8] X. Zhang, Y. Zhou, S. Pei, J. Zhuge, J. Chen, Adversarial examples detection for XSS attacks based on generative adversarial networks, IEEE Access 8 (2020) 10989–10996.
[9] S. Jang, S. Li, Y. Sung, Generative adversarial network for global image-based local image to improve malware classification using convolutional neural network, Appl. Sci. 10 (21) (2020) 7585.
[10] I. Siniosoglou, P. Radoglou-Grammatikis, G. Efstathopoulos, P. Fouliras, P. Sarigiannidis, A unified deep learning anomaly detection and classification approach for smart grid environments, IEEE Trans. Netw. Serv. Manag. 18 (2) (2021) 1137–1151.
[11] M.P. Novaes, L.F. Carvalho, J. Lloret, M.L. Proença Jr., Adversarial deep learning approach detection and defense against DDoS attacks in SDN environments, Future Gener. Comput. Syst. 125 (2021) 156–167.
[12] M. Abdelaty, S. Scott-Hayward, R. Doriguzzi-Corin, D. Siracusa, Gadot: Gan-based adversarial training for robust ddos attack detection, in: 2021 IEEE Conference on Communications and Network Security, CNS, IEEE, 2021, pp. 119–127.
[13] Y. Dong, H. Wang, Y.-D. Yao, A robust adversarial network-based end-to-end communications system with strong generalization ability against adversarial attacks, in: ICC 2022-IEEE International Conference on Communications, IEEE, 2022, pp. 4086–4091.
[14] S.-H. Choi, J.-M. Shin, P. Liu, Y.-H. Choi, ARGAN: Adversarially robust generative adversarial networks for deep neural networks against adversarial examples, IEEE Access 10 (2022) 33602–33615.
[15] X. Zhang, J. Wang, S. Zhu, Dual generative adversarial networks based unknown encryption ransomware attack detection, IEEE Access 10 (2021) 900–913.
[16] J. Shroff, R. Walambe, S.K. Singh, K. Kotecha, Enhanced security against volumetric DDoS attacks using adversarial machine learning, Wirel. Commun. Mob. Comput. 2022 (2022) 1–10.
[17] J. Zhou, Z. Wu, Y. Xue, M. Li, D. Zhou, Network unknown-threat detection based on a generative adversarial network and evolutionary algorithm, Int. J. Intell. Syst. 37 (7) (2022) 4307–4328.
[18] A.-G. Mari, D. Zinca, V. Dobrota, Development of a machine-learning intrusion detection system and testing of its performance using a generative adversarial network, Sensors 23 (3) (2023) 1315.
[19] J. Kipongo, T.G. Swart, E. Esenogho, Artificial intelligence-based intrusion detection and prevention in edge-assisted SDWSN with modified honeycomb structure, IEEE Access (2023).
[20] F.B. Khan, M.H. Durad, A. Khan, F.A. Khan, M. Rizwan, A. Ali, Design and performance analysis of an anti-malware system based on generative adversarial network framework, IEEE Access (2024).
[21] Z. Li, P. Wang, Z. Wang, D.-c. Zhan, FlowGANAnomaly: Flow-based anomaly network intrusion detection with adversarial learning, Chin. J. Electron. 33 (1) (2024) 58–71.
[22] F.M.M. Mokbal, D. Wang, X. Wang, L. Fu, Data augmentation-based conditional Wasserstein generative adversarial network-gradient penalty for XSS attack detection system, PeerJ Comput. Sci. 6 (2020) e328.
[23] X. Liu, X. Di, Q. Ding, W. Liu, H. Qi, J. Li, H. Yang, NADS-RA: network anomaly detection scheme based on feature representation and data augmentation, IEEE Access 8 (2020) 214781–214800.
[24] G. Zhang, X. Wang, R. Li, Y. Song, J. He, J. Lai, Network intrusion detection based on conditional Wasserstein generative adversarial network and cost-sensitive stacked autoencoder, IEEE Access 8 (2020) 190431–190447.
[25] I. Ullah, Q.H. Mahmoud, A framework for anomaly detection in IoT networks using conditional generative adversarial networks, IEEE Access 9 (2021) 165907–165931.
[26] R. Ahsan, W. Shi, X. Ma, W. Lee Croft, A comparative analysis of CGAN-based oversampling for anomaly detection, in: IET Cyber-Physical Systems: Theory & Applications, vol. 7, (1) Wiley Online Library, 2022, pp. 40–50.

[27] Y. Yang, C. Yao, J. Yang, K. Yin, A network security situation element extraction method based on conditional generative adversarial network and transformer, IEEE Access 10 (2022) 107416–107430.

[28] A. Alabrah, A novel study: GAN-based minority class balancing and machine-learning-based network intruder detection using chi-square feature selection, Appl. Sci. 12 (22) (2022) 11662.

[29] J. Xie, S. Li, Y. Zhang, P. Sun, H. Xu, Analysis and detection against network attacks in the overlapping phenomenon of behavior attribute, Comput. Secur. 121 (2022) 102867.

[30] H. Benaddi, M. Jouhari, K. Ibrahimi, J. Ben Othman, E.M. Amhoud, Anomaly detection in industrial IoT using distributional reinforcement learning and generative adversarial networks, Sensors 22 (21) (2022) 8085.

[31] A.S. Dina, A. Siddique, D. Manivannan, Effect of balancing data using synthetic data on the performance of machine learning classifiers for intrusion detection in computer networks, IEEE Access 10 (2022) 96731–96747.

[32] M. Mouyart, G. Medeiros Machado, J.-Y. Jun, A multi-agent intrusion detection system optimized by a deep reinforcement learning approach with a dataset enlarged using a generative model to reduce the bias effect, J. Sens. Actuator Netw. 12 (5) (2023) http://dx.doi.org/10.3390/jsan12050068, URL https://www.mdpi.com/2224-2708/12/5/68.

[33] M. Jamoos, A.M. Mora, M. AlKhanafseh, O. Surakhi, A new data-balancing approach based on generative adversarial network for network intrusion detection system, Electronics 12 (13) (2023) http://dx.doi.org/10.3390/electronics12132851, URL https://www.mdpi.com/2079-9292/12/13/2851.

[34] J. Cui, L. Zong, J. Xie, M. Tang, A novel multi-module integrated intrusion detection system for high-dimensional imbalanced data, Appl. Intell. 53 (1) (2023) 272–288.

[35] M. Adiban, S.M. Siniscalchi, G. Salvi, A step-by-step training method for multi generator GANs with application to anomaly detection and cybersecurity, Neurocomputing 537 (2023) 296–308.

[36] G.-C. Lee, J.-H. Li, Z.-Y. Li, A wasserstein generative adversarial network–gradient penalty-based model with imbalanced data enhancement for network intrusion detection, Appl. Sci. 13 (14) (2023) http://dx.doi.org/10.3390/app13148132, URL https://www.mdpi.com/2076-3417/13/14/8132.

[37] C. Park, J. Lee, Y. Kim, J.-G. Park, H. Kim, D. Hong, An enhanced AI-based network intrusion detection system using generative adversarial networks, IEEE Internet Things J. 10 (3) (2022) 2330–2345.

[38] Y.N. Rao, K. Suresh Babu, An imbalanced generative adversarial network-based approach for network intrusion detection in an imbalanced dataset, Sensors 23 (1) (2023) 550.

[39] A.S. Barkah, S.R. Selamat, Z.Z. Abidin, R. Wahyudi, Data generative model to detect the anomalies for IDS imbalance CICIDS2017 dataset, TEM J. 12 (1) (2023).

[40] C. Wang, D. Xu, Z. Li, D. Niyato, Effective intrusion detection in highly imbalanced IoT networks with lightweight S2CGAN-IDS, IEEE Internet Things J. (2023).

[41] S. Wang, H. Chen, L. Ding, H. Sui, J. Ding, GAN-SR anomaly detection model based on imbalanced data, IEICE Trans. Inf. Syst. 106 (7) (2023) 1209–1218.

[42] J. Zhai, P. Lin, Y. Cui, L. Xu, M. Liu, GraphCWGAN-GP: A novel data augmenting approach for imbalanced encrypted traffic classification, CMES Comput. Model. Eng. Sci. 136 (2) (2023).

[43] K.S. Babu, Y.N. Rao, MCGAN: modified conditional generative adversarial network (MCGAN) for class imbalance problems in network intrusion detection system, Appl. Sci. 13 (4) (2023) 2576.

[44] J. He, X. Wang, Y. Song, Q. Xiang, C. Chen, Network intrusion detection based on conditional wasserstein variational autoencoder with generative adversarial network and one-dimensional convolutional neural networks, Appl. Intell. 53 (10) (2023) 12416–12436.

[45] Z. Cai, H. Du, H. Wang, J. Zhang, Y. Si, P. Li, One-dimensional convolutional wasserstein generative adversarial network based intrusion detection method for industrial control systems, Electronics 12 (22) (2023) 4653.

[46] B. Xuan, J. Li, Y. Song, SFCWGAN-BITCN with sequential features for malware detection, Appl. Sci. 13 (4) (2023) 2079.

[47] K.T. Chui, B.B. Gupta, P. Chaurasia, V. Arya, A. Almomani, W. Alhalabi, Three-stage data generation algorithm for multiclass network intrusion detection with highly imbalanced dataset, Int. J. Intell. Netw. 4 (2023) 202–210.

[48] M. Zaman, D. Upadhyay, C.-H. Lung, Validation of a machine learning-based IDS design framework using ORNL datasets for power system with SCADA, IEEE Access 11 (2023) 118414–118426.

[49] Q. Zhou, Z. Wang, A network intrusion detection method for information systems using federated learning and improved transformer, Int. J. Semant. Web Inf. Syst. (IJSWIS) 20 (1) (2024) 1–20.

[50] J. Saikam, K. Ch, EESNN: Hybrid deep learning empowered spatial-temporal features for network intrusion detection system, IEEE Access (2024).

[51] C. Sweet, S. Moskal, S.J. Yang, On the variety and veracity of cyber intrusion alerts synthesized by generative adversarial networks, ACM Trans. Manag. Inf. Syst. (TMIS) 11 (4) (2020) 1–21.

[52] M. Mimura, Using fake text vectors to improve the sensitivity of minority class for macro malware detection, J. Inf. Secur. Appl. 54 (2020) 102600.

[53] J. Chen, X. Zhang, R. Zhang, C. Wang, L. Liu, De-pois: An attack-agnostic defense against data poisoning attacks, IEEE Trans. Inf. Forensics Secur. 16 (2021) 3412–3425.

[54] M. Kang, H. Kim, S. Lee, S. Han, Resilience against adversarial examples: Data-augmentation exploiting generative adversarial networks, KSII Trans. Internet & Inf. Syst. 15 (11) (2021).

[55] A. Liu, Y. Wang, T. Li, SFE-GACN: A novel unknown attack detection under insufficient data via intra categories generation in embedding space, Comput. Secur. 105 (2021) 102262.

[56] M.G. Sahakian, S. Musuvathy, J. Thorpe, S. Verzi, E. Vugrin, M. Dykstra, Threat data generation for space systems, in: 2021 IEEE Space Computing Conference, SCC, IEEE, 2021, pp. 100–109.

[57] K.-H. Le, M.-H. Nguyen, T.-D. Tran, N.-D. Tran, IMIDS: An intelligent intrusion detection system against cyber threats in IoT, Electronics 11 (4) (2022) 524.

[58] B.A. Alabsi, M. Anbar, S.D.A. Rihan, Conditional tabular generative adversarial based intrusion detection system for detecting ddos and dos attacks on the internet of things networks, Sensors 23 (12) (2023) 5644.

[59] S. Alabdulwahab, Y.-T. Kim, A. Seo, Y. Son, Generating synthetic dataset for ML-based IDS using CTGAN and feature selection to protect smart IoT environments, Appl. Sci. 13 (19) (2023) 10951.

[60] S. Fathi-Kazerooni, R. Rojas-Cessa, GAN tunnel: Network traffic steganography by using GANs to counter internet traffic classifiers, Ieee Access 8 (2020) 125345–125359.

[61] M. Romanelli, K. Chatzikokolakis, C. Palamidessi, Optimal obfuscation mechanisms via machine learning, in: CSF, 2020, pp. 153–168.

[62] S. Desai, N. Sabar, R. Alhadad, A. Mahmood, N. Chilamkurti, Mitigating Consumer Privacy Breach in Smart Grid Using Obfuscation-Based Generative Adversarial Network, La Trobe, 2022.

[63] S. Tyagi, S. Mittal, Sampling approaches for imbalanced data classification problem in machine learning, in: Proceedings of ICRIC 2019: Recent Innovations in Computing, Springer, 2020, pp. 209–221.

[64] J. Liu, Y. Tian, R. Zhang, Y. Sun, C. Wang, A two-stage generative adversarial networks with semantic content constraints for adversarial example generation, IEEE Access 8 (2020) 205766–205777.

[65] R. Chauhan, S.S. Heydari, Polymorphic adversarial ddos attack on IDS using GAN, in: 2020 International Symposium on Networks, Computers and Communications, ISNCC, IEEE, 2020, pp. 1–6.

[66] S. Zhao, J. Li, J. Wang, Z. Zhang, L. Zhu, Y. Zhang, Attackgan: Adversarial attack against black-box ids using generative adversarial networks, Procedia Comput. Sci. 187 (2021) 128–133.

[67] E. Alshahrani, D. Alghazzawi, R. Alotaibi, O. Rabie, Adversarial attacks against supervised machine learning based network intrusion detection systems, Plos One 17 (10) (2022) e0275971.

[68] R.M. Arif, M. Aslam, S. Al-Otaibi, A.M. Martinez-Enriquez, T. Saba, S.A. Bahaj, A. Rehman, A deep reinforcement learning framework to evade black-box machine learning based IoT malware detectors using GAN-generated influential features, IEEE Access 11 (2023) 133717–133729, http://dx.doi.org/10.1109/ACCESS.2023.3334645.

[69] S. Aldhaheri, A. Alhuzali, SGAN-IDS: Self-attention-based generative adversarial network against intrusion detection systems, Sensors 23 (18) (2023) 7796.

[70] X. Ma, K. Yang, C. Zhang, H. Li, X. Zheng, Physical adversarial attack in artificial intelligence of things, IET Commun. (2023).

[71] Z. Liu, J. Hu, Y. Liu, K. Roy, X. Yuan, J. Xu, Anomaly-based intrusion on IoT networks using AIGAN-a generative adversarial network, IEEE Access (2023).

[72] S.A. Khowaja, P. Khuwaja, K. Dev, A. Antonopoulos, Spin: Simulated poisoning and inversion network for federated learning-based 6g vehicular networks, in: ICC 2023-IEEE International Conference on Communications, IEEE, 2023, pp. 6205–6210.

[73] Y. Shi, K. Davaslioglu, Y.E. Sagduyu, Generative adversarial network in the air: Deep adversarial learning for wireless signal spoofing, IEEE Trans. Cogn. Commun. Netw. 7 (1) (2020) 294–303.

[74] K. Zerhouni, G.S. Gaba, M. Hedabou, T. Maksymyuk, A. Gurtov, E.M. Amhoud, GAN-based evasion attack in filtered multicarrier waveforms systems, IEEE Trans. Mach. Learn. Commun. and Netw. (2024).

[75] E.C.P. Neto, S. Dadkhah, R. Ferreira, A. Zohourian, R. Lu, A.A. Ghorbani, CICIoT2023: A real-time dataset and benchmark for large-scale attacks in IoT environment, Sensors 23 (13) (2023) http://dx.doi.org/10.3390/s23135941, URL https://www.mdpi.com/1424-8220/23/13/5941.

[76] R. Mills, LUFlow Network Intrusion Detection Data Set, Kaggle, 2024, http://dx.doi.org/10.34740/KAGGLE/DSV/8106645, URL https://www.kaggle.com/dsv/8106645.

[77] I. Crespo, A. Campazas, SQL Injection Attack Netflow, Zenodo, 2022, http://dx.doi.org/10.5281/zenodo.6907252.



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